

The Information Society: Accessible to All?

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Introduction

This chapter analyses the most common barriers to accessing ICT (Information and Communication Technologies)-based products and services faced by people with disabilities, as well as the strategies currently considered most effective in addressing these obstacles.

In the early 20th century, Spanish society underwent an unprecedented revolution brought about by the expansion of the telephone network. This technological revolution brought with it obvious changes to the landscape, such as the first telephone lines connecting major towns, the first public telephone booths, telephone exchanges, and so on.

Nowadays, the revolution in communications, which began in the late 20th century, is experiencing a new surge. However, unlike a century ago, this transformation is less obvious, at least in terms of the physical landscape, as communication infrastructure is no longer as visible. Information flows through fibre-optic cables running beneath the streets or through radio waves transmitted via relay stations.

Data can even be transmitted through our bodies. The increasing convergence of previously distinct fields, such as biology and information technology, has prompted specialists to examine questions regarding the extent to which future individuals might be defined as human, based on the proportion of their bodies replaced by “smart” prosthetic devices, particularly those possessing operational autonomy.

The current information society is marked by a number of distinctive features:

- Growing communication networks: Not only the Internet (the communication network par excellence) but also the number of devices connected to it is expanding exponentially, and its infrastructure is capable of supporting seemingly limitless growth.
- Paradigm shift regarding the value of information: The wealth of information stored on the internet does not stem from its mere existence, but from the fact that it can be shared by an unlimited number of people, and that some of them are also sharing personal data (such as identity, location, preferences, etc.), knowingly or unknowingly.
- Citizen-government relations are changing: People are taking advantage of the widespread availability of communication services to organise themselves for various reasons and with different outcomes. Examples of this include mass demonstrations for political reasons, or even the coordination of violent protests.

Public and private organisations are providing an increasing number of services that users can access via ICT-based devices. Furthermore, some services are no longer provided face-to-face. Instead, they are offered exclusively through digital channels, such as websites, automated telephone services, vending machines, information kiosks, and so on.

In reality, a significant number of people are unable to access basic ICT services, such as making a phone call or searching for information online, as well as services that are now being provided via ICT. This may be due to the following reasons:

- Limited geographical availability. Electronic services are simply not provided in certain parts of the world. Mobile and fixed-line telephone

coverage, for example, is significantly slower to be rolled out in rural areas with low population density compared to urban areas, where services are provided more quickly.

- **Gender issues:** Women have traditionally faced greater difficulties in accessing these services in countries where, due to culture or religion, they are marginalised, which limits their literacy levels, an essential condition for accessing ICTs.
- **Age:** Age alone does not prevent people from accessing technology. However, elderly people tend to make less use of new technologies than younger and middle-aged people. This trend is likely to change in the future. Nevertheless, radical changes in the technological landscape will continue to create barriers between generations unless addressed through good design practices.
- **Socio-economic disparities:** A significant share of the world's population lives below the poverty line and has limited access to ICT. Developing countries tend to pay a higher relative cost for these services, such as internet access, than more developed countries.
- **Disability:** People with disabilities encounter many types of barriers when accessing products and services in the information society.

The first three of the five causes of exclusion mentioned above tend to become less significant as societies become more developed. As for the last two (socio-economic disparities and disability), these are structural limitations that must be addressed through more specific strategies. The lack of access to the Information Society is sometimes referred to as the digital divide. [Table 1.1](#) lists the causes underlying the ongoing digital divide, based on a study conducted in Europe.

Table 1.1. What factors contribute to the ongoing digital divide? (European Commission - Information Society and Media, 2009)

Causes of the digital divide	Percentage of responses
The lack of accessibility and usability remains a major barrier	68.90%
Relevance: People are not aware of the possibilities and potential benefits, or are not interested	59.20%
Cost: The cost of ICT services is a barrier	55.30%
Availability: Gaining access to ICT remains a problem	30.10%
Compatibility between ICT services and devices is often a problem	26.20%
Others	19.40%
Physical accessibility is inadequate	13.60%

The Report on the public consultation on European e-Inclusion Policy (European Commission – Information Society and Media, 2009) found that 40% of the European population were not fully benefiting from the Information Society in 2009. This is a serious issue because people who are unable to use services in their digital form risk being unable to use them at all. This is particularly true as

services are increasingly being phased out in their traditional or “face-to-face” forms. In a society where an increasing number of basic services (health, education, social services, etc.) are provided via ICT, many people stand to be excluded from the Information Society and, consequently, find their human rights violated.

The concept of disability

The following section will focus on one of the reasons for exclusion from the information society: disability. The definition of this term has changed over time. In 1980, the WHO (World Health Organization) defined disability as a limitation in an individual’s ability to perform a specific activity or to participate in society in general. According to this definition, an impairment, caused by a disease, results in a disability (WHO, 1980), as illustrated in [Figure 1.1](#).

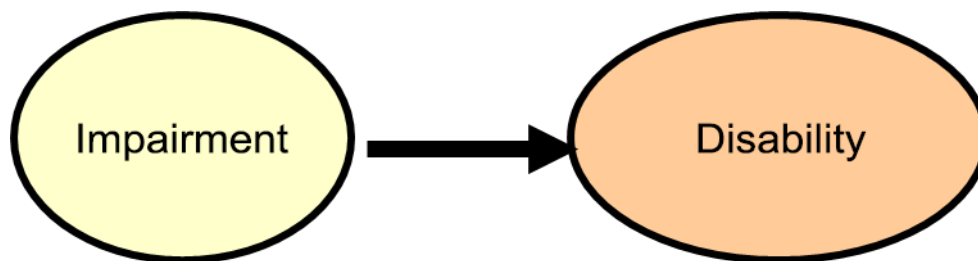


Figure 1.1. The link between Impairment and Disability, according to the WHO in 1980

In 2001, the WHO added the following to the previous definition: disability is characterised by the result of a complex interaction between a person’s health status and personal and environmental factors, which represent the circumstances in which the person lives (WHO, 2001), as shown in [Figure 1.2](#).

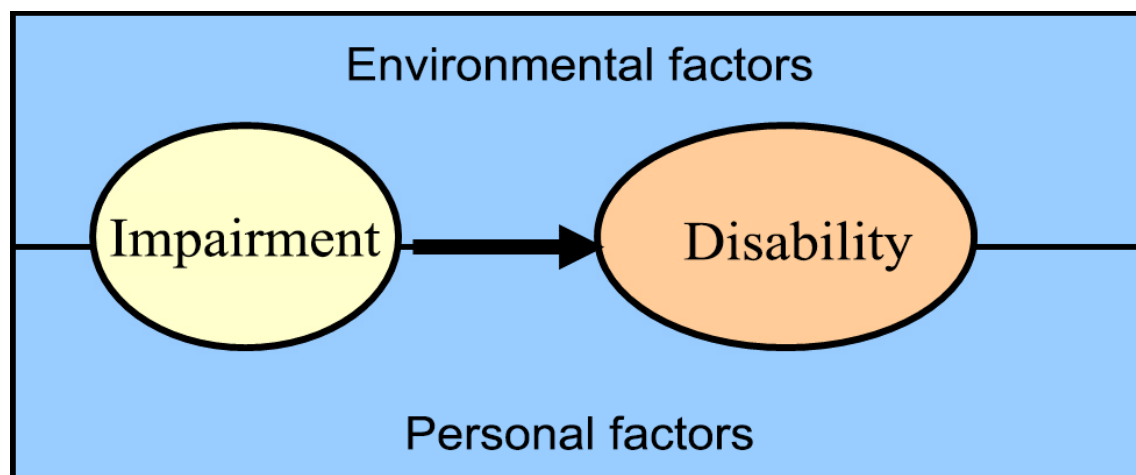


Figure 1.2. The link between Impairment and Disability, according to the WHO in 2001

According to the WHO (2001):

- Environmental factors refer to the physical, social and attitudinal environment in which a person lives and conducts their life. These factors are beyond the individual's control and can have a negative or positive

influence on the person's performance/achievement as a member of society, on their capabilities, or on their bodily structures and functions.

- Personal factors include an individual's life history and aspects that are not part of a health condition or state of health. These factors may include gender, race, age, other health conditions, lifestyle, habits, coping strategies, social background, education, occupation, current and past experiences (past and present life events), behavioural patterns and personality style, psychological aspects and other characteristics, all or some of which play a role in any level of disability.

This new WHO definition marked a conceptual departure from the traditional understanding of disability. Unlike previous interpretations, it acknowledges that environmental factors interact with a person's body structures and functions, as well as with their activities and participation. Therefore, a comprehensive analysis is carried out regarding the influence of the environment on individuals, concluding that disability is the result of an interaction between people with impairments and a world (environment) that is not adequately designed for them.

In summary, whereas the World Health Organization in 1980 emphasised the effects stemming from diseases and health conditions (such as blindness precluding the ability to read a computer screen), its 2001 framework recognises that disability is not exclusively dictated by an individual's health status. Rather, it is also shaped by personal attributes and contextual factors. Due to the person-environment interaction, various environments can have diverse impacts on the same individual with a given health condition. An obstacle-filled environment will restrict the individual's performance; other environments where no obstacles exist, or where support is available, can enhance that individual's fulfilment.

The concept of functional diversity

The terms "impairment" and "disability" carry negative connotations. According to the English Dictionary, the prefix "dis-" or "im-" denotes negation or opposition. Alternative concepts have emerged, such as "organic difference" instead of "impairment" and "functional diversity" instead of "disability" (Palacios & Románach, 2006). According to the concept of functional diversity, every human being functions differently, and therefore all human beings are diverse. Palacios and Románach propose using "people discriminated against for functional diversity" instead of "people with disabilities," noting that such discrimination is a human rights violation.

Here are some examples to clarify these ideas:

- Someone with a spinal cord injury (referred to as an "impairment" by the WHO, or an "organic difference" by Palacios and Románach) who cannot move around independently may struggle to "go to the top floor of the building" if the only way to do so is by climbing stairs. However, if that person's environment is suitable (a ramp and a wheelchair, in this case), they can perfectly well perform that task.
- Deaf people can access the audio content of audiovisual material if it is available in the alternative format best suited to them (which for some deaf people will be subtitles for the deaf or hard of hearing, and for others, sign language interpretation).

- A visually impaired person can read a book if it is available in accessible digital format, so that he/she can listen to its contents and navigate through them.

Functional diversity is inherent to human beings. Some human beings:

- can see little or nothing,
- can hear little or nothing,
- have limited mobility or difficulty using their upper limbs,
- are above or below average height,
- have difficulty expressing themselves verbally,
- have problems learning or remembering.

Even the same person can experience changes in how they function over the course of their lifetime. Humans are highly dependent at birth, needing assistance for movement, nourishment, and survival. As we develop, our ability to communicate with others and use objects, and so on, tends to become increasingly independent.

That independence can be challenged by moments of “temporary” dependence. For instance, this might happen when we visit a country where we can't read or speak the language, find ourselves in a loud setting that makes phone calls difficult, or suffer an arm injury that keeps us from using a computer mouse, among other situations. However, as we reach a certain stage in life, every human being's abilities begin to decline, including our eyesight, our ability to move about, to remember and to understand. As a result, anyone who is fortunate enough to enjoy a long life is very likely to develop one or more legally recognised disabilities. The concepts that define human diversity and its associated challenges will likely continue to evolve in the future. Hopefully, this development will lead to changes in society's prevailing ideas, and consequently to debates, political and legislative reform, and a shift in social reality.

Personal autonomy, accessibility and personal assistance

The concept of personal autonomy and how it relates to quality of life is described by García Alonso (2003). Every member of society aims to enjoy a certain standard of living. This requires a series of needs to be met. There are two essential needs: Optimal physical health and optimal personal autonomy.

The elements necessary to achieve personal autonomy are accessibility and personal assistance.

One of these elements, accessibility, is a key part of our course and will be studied in detail. This can be accomplished through two complementary strategies:

- Design for All
- Assistive Technology.

Personal assistance is no less important. Without delving too deeply into the concept, even in an accessible environment, i.e. one with products designed for everyone and the necessary assistive devices, some people still need help from another person to carry out specific (and sometimes essential) tasks, such as connecting to a device, reaching equipment, or positioning assistive devices correctly, etc. That person is known as a personal assistant.

Consider a person using their computer to write a text document, aided by a voice recognition system. Before performing that task, that person has got out of bed, washed, had breakfast, sat down at the computer and switched it on. Some people

with limited mobility will have needed personal assistance to carry out these preliminary tasks.

Discrimination based on functional diversity in the Information Society

A significant proportion of citizens find it difficult to use ICT products and services due to their functional diversity. These cases can be described as discrimination, given that these services are essential to their quality of life. People in this situation will struggle more than others to access basic services and, consequently, to enjoy full citizenship.

The following are some typical scenarios where users may find it difficult to use e-services or e-products:

- Wheelchair users who find it difficult to use an ATM because the architectural design of the surroundings makes it impossible for them to even get close to the screen or the controls.
- Blind people using a screen reader who cannot access their university's websites because they cannot use the keyboard to access basic navigation functions, such as the list of links on the page.
- People with cognitive impairments who cannot access public resources that might be useful to them, simply because these are provided via difficult-to-understand websites and texts.
- Deaf people who cannot access the 112 emergency services, which lack adapted communication mechanisms.

According to the Report on the public consultation on European e-Inclusion Policy, 2009 (European Commission – Information Society and Media, 2009): “Access to the Information Society is a fundamental right that enables older people and people with disabilities to live independently and participate fully in the Information Society, on an equal footing with their fellow citizens. ICTs represent a real opportunity to combat isolation and social exclusion. They play a key role in the daily lives of older people and people with disabilities, who use them to a greater extent than the general population. Furthermore, ICTs facilitate access to education, work and entertainment. They enable older people to be included in all aspects of social life”.

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